

DEPARTMENT OF COMPUTER SCIENCE

Mid Semester Examination (Fall-2024)

Course Code: CS-201
 Course Title: Data Structures
 Total Marks: 25
 Date: 02-12-2024

Name: _____
 Register No: _____
 Discipline/Semester: CS/CS/AI/CYS 3rd
 Time Allowed: 90minutes

NOTE:

- *Attempt all the questions. Understanding question is part of exam.
- *Maintain the order of the questions same as question paper in your answer sheet.
- Return question paper with answer sheets.

Section I: [CLO1-GA2] [C2] [2x3+4 = 10 Marks]

Q. 1: Briefly explain the following:

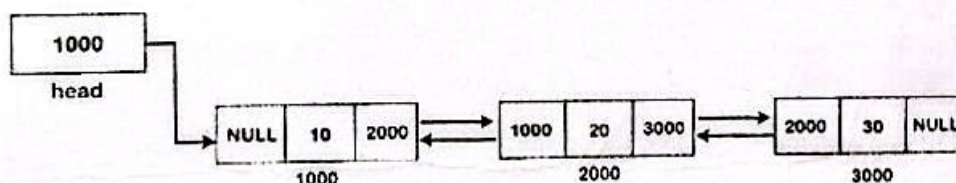
- i. Linear and non-linear data structure
- ii. Types of asymptotic notations

Q. 2: A 2D array GRID has dimensions 300×400 , stored in row-major order. Each element requires 1 byte of memory. The base address of the array is 8000. The lower bound of array is one. Compute the memory address of the element at GRID [100][200].

Section II: [CLO2-GA4] [C3] [5+5+5=15 marks]

Q. 3: Consider you are developing a student management system where the students' records are stored using singly linked list. Each node in the list represents a student, containing fields like the student's ID. Write a pseudocode to implement a functionality to delete a specific student record from the list, identified by the student's unique ID.

Q. 4: Consider the following doubly linked list and construct pseudocode to insert 5 at address 500 by considering the order of addressing scheme. Also show the pictorial representation of updated list.



Q. 5: Consider the following infix expression and compute its equivalent postfix notation using stack.

$$(A * B) + (C - (D / E ^ F) * G) * H$$

"Transforming our world begins with simple actions—let's save energy today for a better tomorrow."

Course Code: GE-154
Course Title: Calculus and Analytic
Geometry (QR-II)
Total Marks: 30
Date: 06-12-2024

Name: _____
Registration No. _____
Discipline/Semester: BS CS/AM/CYC-III
Time Allowed: 1 hr. 30 mins.

**NOTE: *Attempt all the questions. Understanding question is part of exam.
*Maintain the order of the questions same as question paper in your answer sheet.**

Section I: [CLO1-GA1] [C1] [7+5 = 12 marks]

Q. 1: Define the tangent and normal line and find the equation of tangent and normal lines for
 $f(x) = 2x^3 - 3x + 7$ at (1,6).

Q. 2: Show that given function $f(x) = \begin{cases} \frac{x-4}{\sqrt{x}-2} & \text{if } x \neq 4 \\ 4 & \text{if } x = 4 \end{cases}$
is continuous at $x = 4$.

Section II: [CLO2-GA2] [C2] [2x2.5+5 = 10 marks]

Q. 3: Differentiate the following:

(i) $y = \left(\frac{3x^2-5}{2x^2+7}\right)^2$ (ii) $y = \sin(x + e^x)$

Q. 4: Determine $\frac{dy}{dx}$, if $x^2y^4 + 3x = 8y^3 + 3$.

Section III: [CLO3-GA3] [C3] [5+3 = 08 marks]

Q. 5: Determine whether the function f satisfies the hypothesis of the Mean Value Theorem on the indicated interval $[a,b]$ and if so, calculate all numbers c in (a,b)
 $f(x) = x^3 - 2x^2 + x + 3, [-1,1]$.

Q. 6: Find the absolute maximum and minimum of f on the indicated closed interval.
 $f(x) = x^4 - 5x^2 + 4; [0,2]$

UW

DEPARTMENT OF COMPUTER SCIENCE

MID Semester Examination (Fall-2024)

Course Code: CS-251

Course Title: Introduction to Information Security

Total Marks: 30

Date: 04-12-24

Name _____

Registration N _____

Discipline/Semester: BSCS/BSAI/BSUY 3rd

Time Allowed: 90 minutes

NOTE: *Maintain the order of the questions same as question paper in your answer sheet.

Section 01: [CLO1, GA2] [C2] [9+5 = 14 marks]

Q1. Answer the following questions

- Describe* the three aspects of security?
- Discuss* different types of Active and Passive attacks?
- Explain* the key steps involved in the RSA algorithm, and how do they ensure secure encryption and decryption?

Q2. *Describe* the five ingredients of the Symmetric Cryptography with the help of diagram and explain how encryption is less secure in symmetric cryptography in terms of key management.

Section 02: [CLO 2, GA 3, C3] [5 + 6 + 5 = 16 marks]

Q3. **Encrypt** the message Program Name (e.g BS Computer Science) using Rail Fence Technique and **Decrypt** the encrypted message to confirm that it matches the original message with depth/key=4.

Q4. **Apply** the Vernam cipher technique on the given plaintext "ARTIFICIAL" with the key "BLOCKCHAIN" for encryption and decryption using XoR operation.

Q5. **Encrypt** the message "SAFE MESSAGES" using the Hill Cipher Technique to convert the plain text into cipher text by using the key matrix $\begin{pmatrix} 2 & 8 \\ 5 & 7 \end{pmatrix}$.

*****GOOD LUCK😊*****

Course Code: CS-241
Course Title: Computer Networks
Total Marks: 30
Date: 03rd December, 2024

Name: _____
Registration No: _____
Discipline/Seme: _____
Time Allowed: 90 minutes

NOTE:

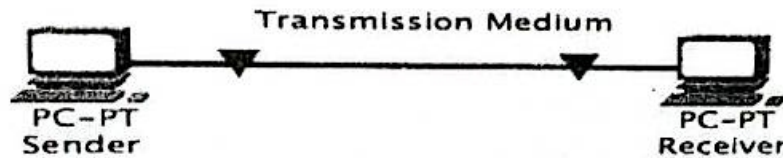
- * Attempt all the questions.
- * Return Question Paper along with answer sheet.
- * Maintain the order of the questions same as question paper in your answer sheet.

Section 1: [CLO1-GA1] [C2] [[06 + 05 + 05 = 16 marks]

Q1. Explain the following:

- a. Data link Layer and Transport Layer
- b. Client Server and P2P Architecture
- c. FECN and BECN

Q2. Suppose a sender wants to communicate with a receiver through a transmission medium. Describe with the help of a diagram, how encapsulation and decapsulation processes occur across the OSI model layers.



Q3. Explain the HTTP protocol and how communication occurs between the client and server using HTTP.

Section 2: [CLO2-GA2] [C3] [[05 marks]

Q4. Illustrate how data flows in circuit switching versus packet switching during a communication session.

Section 3: [CLO3-GA3] [C4] [[03 + 03 + 03 = 09 marks]

Q5. In a TCP Slow Start phase with an initial congestion window (cwnd) size of 1 segment and a threshold of 16 segments, examine the following with the help of diagram:

- a. The total number of rounds needed for the congestion window size to reach or exceed the threshold.
- b. The total number of acknowledgments received during this period (consider that all packets sent in one round are acknowledged before the next round begins).
- c. Using a diagram, illustrate how a connection is established in the TCP protocol.

Small changes can lead to big impacts—reduce, reuse, and save energy for a brighter future

UW

DEPARTMENT OF COMPUTER SCIENCE
MID Semester Examination (Fall 2024)

Course Code: CS-202

Course Title: Artificial Intelligence

Total Marks: 30

Date: 05-12-2024

Name: _____

Registration No: _____

Discipline/Semester: CS CS 377 / AI 3 / 2024

Time Allowed: 90 minutes

NOTE:

*Attempt all the questions.

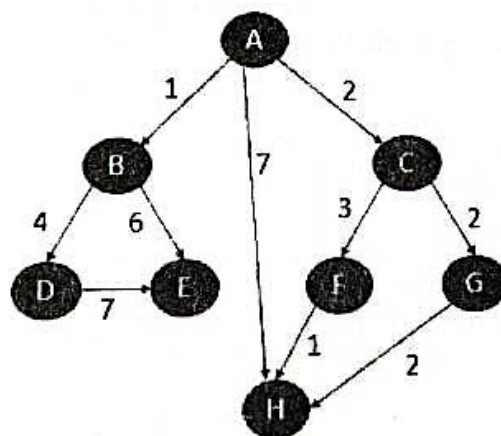
*Maintain the order of the questions same as question paper in your answer sheet.

Section 1: [CLO1-GA1] [C2] [6 + 4 = 10 marks]**Q. 1: Define the following and answer must be brief and precise:****[3 * 2 = 6 Marks]**

- Knowledge-base
- Inference engine
- Episodic vs. sequential Environment

Q. 2: Demonstrate the working of Goal-Based agent with the help of a diagram.**[4 Marks]****Section 2: [CLO3-GA4] [C4] [2.5 + 2.5 + 3 = 8 Marks]****Q. 3: Examine the scenario of designing an "Intelligent Personal Assistant System for Healthcare" and for this scenario answer the following questions:**

- Describe the PEAS components for the scenario.
- Classify the type of environment
- Problem description components

Section 3: [CLO2] [GA3] [C3] [7 + 5 = 12 Marks]**Q. 4: Apply the following search algorithms considering the graph given below:**

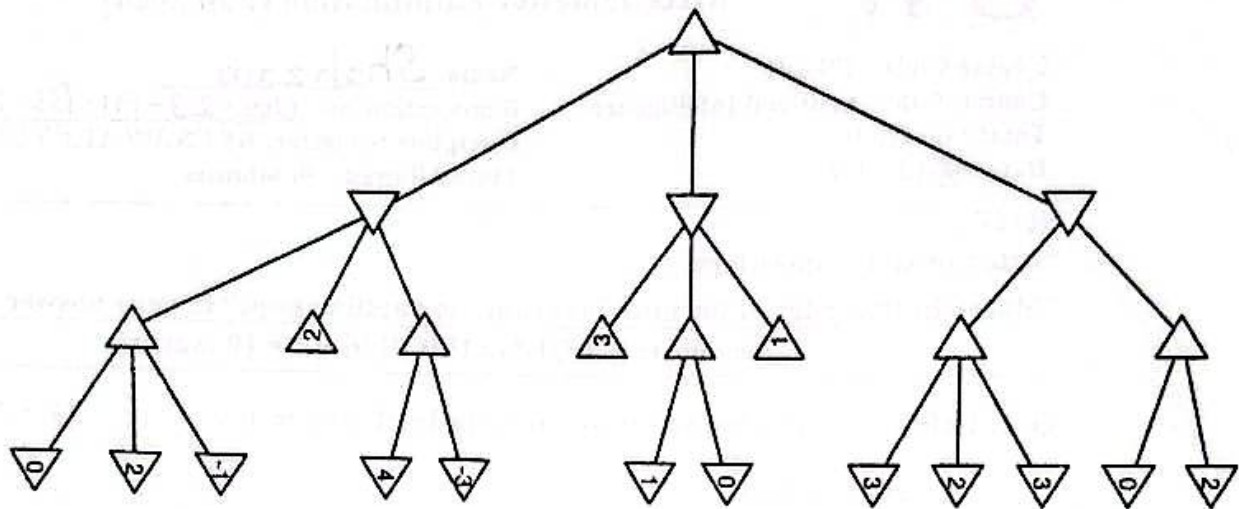
NODES	HEURISTICS
A	5
B	3
C	4
D	2
E	6
F	3
G	1
H	0

- Breadth-First Search [3 Marks]
- A* Algorithm [4 Marks]

P.T.O

Q. 5: Apply the alpha-beta pruning algorithms on the graph given below:

[5 Marks]



*****GOOD LUCK*****